

An Introduction to $\text{\LaTeX}$

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OIA $\text{\LaTeX}$ workshop

Outline

Introduction

What is $\text{\LaTeX}$ and why?

Using $\text{\LaTeX}$

The basics

The basics of formatting

Document structure

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Why L^AT_EX?

If you merely want to produce a passably good document — something acceptable and basically readable but not really beautiful — a simpler system will usually suffice.

With T_EX the goal is to produce the finest quality.

Donald E. Knuth

- ▶ Program for typesetting technical text
 - ▶ Specialised in formulas
 - ▶ Professional typographic design
 - ▶ Freely available
 - ▶ Stable
- ▶ File format based on ASCII
 - ▶ Easy to send by e-mail
 - ▶ Works on all systems
- ▶ Easily extendable through macros

- ▶ Macro package for T_EX
 - ▶ Freely available
 - ▶ Stable
- ▶ Logical structure
- ▶ Widely used in the exact sciences
 - ▶ Publishers require the use of their own style files with macros for their own layout

L^AT_EX workflow

- ▶ Create ASCII file(s) with L^AT_EX commands
- ▶ Create a DVI file from this with `latex` (Device Independent)
- ▶ Turn the DVI file into a PostScript file with `dvips` or a PDF with `dvi2pdf`, or
- ▶ Create a PDF file directly with `pdflatex`
- ▶ Output can be viewed with
 - ▶ DVI: `xdvi`, `yap`
 - ▶ PS: Ghostscript
 - ▶ PDF: Acrobat Reader

L^AT_EX tools

- ▶ There are several T_EX distributions
- ▶ Any ASCII editor is fine
- ▶ L^AT_EX is not WYSIWYG
- ▶ There are tools to make working with L^AT_EX easier:
 - ▶ WYSIWYG editor (LyX)
 - ▶ L^AT_EX shells or IDEs (WinShell, Context, TeXnicCenter)
 - ▶ Editors with syntax highlighting (Emacs)

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Basic L^AT_EX document

Start with a document class such as `article`.

Basic L^AT_EX document

A minimal file also needs `\begin{document}` and `\end{document}`.

Basic L^AT_EX document

Add packages before the document starts, for example
`\usepackage[english,dutch]{babel}`.

Basic L^AT_EX document

You can then start structuring the content with sections.

Basic L^AT_EX document

Typical sectioning commands are `\section`, `\subsection`, and `\subsubsection`.

Basic L^AT_EX document

Table of contents entries are generated automatically from the sectioning commands.

Basic L^AT_EX document

- ▶ `\section{Introduction}`
- ▶ `\section{An interesting section}`
- ▶ `\section{Summary}`
- ▶ `\tableofcontents` can be used to print the outline

Text formatting

- ▶ $\text{\LaTeX}$ automatically handles:
 - ▶ automatic numbering of chapters, sections, and so on
 - ▶ correct spacing between letters, words, and sentences
 - ▶ ligatures
 - ▶ word breaking (with `\selectlanguage{english}`)
 - ▶ justification of paragraphs and pages

Input

- ▶ Forbidden characters include `# $ % ^ & _ { }` and the backslash.
- ▶ Literal output is usually written with `\verb` or the `verbatim` environment.
- ▶ Ready-made strings:
 - ▶ `\today` today
 - ▶ `\TeX` T_EX
 - ▶ `\LaTeX` L^AT_EX

Special characters

- ▶ Quotes:
 - ▶ “click ‘Start’ to begin” becomes typographic quotes in $\text{\LaTeX}$
- ▶ Hyphens:
 - ▶ X-rated
 - ▶ page 2–24
 - ▶ yes — or is it no
 - ▶ π and $-\pi$
- ▶ Ellipsis: ...
- ▶ Accents and special symbols: ö is o-umlaut, ô is o-circumflex, ō is o-macron, Æ is AE

Fonts

<code>\textrm{...}</code>	roman
<code>\textsf{...}</code>	sans serif
<code>\texttt{...}</code>	typewriter
<code>\textmd{...}</code>	medium
<code>\textbf{...}</code>	bold
<code>\textup{...}</code>	upright
<code>\textit{...}</code>	italic
<code>\textsl{...}</code>	slanted
<code>\textsc{...}</code>	small caps
<code>\emph{...}</code>	emphasized
<code>\textnormal{...}</code>	document font

Special use of fonts

- ▶ Emphasis
 - ▶ text text
 - ▶ *emphasis* emphasis
- ▶ Selection of fonts and sizes
 - ▶ *You can also emphasize something if the text is italic*
 - ▶ or in sans-serif style
 - ▶ or in typewriter style
 - ▶ **And also if it is bold!**

Font sizes

<code>\tiny</code>	tiny
<code>\scriptsize</code>	extremely small
<code>\footnotesize</code>	very small
<code>\small</code>	small
<code>\normalsize</code>	normal
<code>\large</code>	large
<code>\Large</code>	larger
<code>\LARGE</code>	even larger
<code>\huge</code>	huge
<code>\Huge</code>	largest

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Titles, chapters, sections, and so on

- ▶ Title: `\title`, `\author`, and `\maketitle`
- ▶ Chapters, sections, and so on
 - ▶ In the article document class: `\section`, `\subsection`, `\subsubsection`, `\appendix`
 - ▶ In the book and report document classes: `\chapter`, `\part`

References

Input

Labels let you refer back to a section and its page number later in the document.

2. Preparation

And here comes the text of this paragraph....

3. Solution

In section 2 on page 10 we wrote ...

Bibliography

Input

Citations are inserted with a citation command and a bibliography environment.

Output

D. Knuth [1] mentioned this ...

Bibliography

1 Knuth, Donald E.: The TeXBook (1984), Addison-Wesley Publishing Company, Reading (USA). 480 p.

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Lists: enumerate, itemize, description

1. first item
 2. second item
- ▶ an item in itemize
 - and another one!
- high a description
- low final example

Lists: enumerate, itemize, description (nested)

1. first item

- ▶ a subitem
 - and another one!

2. second item

high description
low and another one

Tables

Input

A tabular environment aligns numbers and labels into columns.

Expenses	
1000	travel
3700	lodging
800	conference
5500	total

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Math mode

- ▶ $\text{\LaTeX}$ has a special mode for working with mathematical formulas:
 - ▶ Inline in text: between \$ signs
 - ▶ Standalone unnumbered formula: between `\[` and `\]` (or between `\begin{displaymath}` and `\end{displaymath}`)
 - ▶ Standalone numbered formula: between `\begin{equation}` and `\end{equation}`

Examples

Pythagoras proved that $a^2 = b^2 + c^2$. Moreover,

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k^2} = \frac{\pi^2}{6}. \quad (1)$$

In equation (1) we see ...

Examples

Pythagoras proved that $a^2 = b^2 + c^2$. Moreover,

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k^2} = \frac{\pi^2}{6}.$$

In equation (1) we see ...

More ways to typeset mathematical formulas

$$y = \begin{cases} a & \text{if } d > c \\ b + x & \text{in the morning} \\ \text{all day} \end{cases}$$

$$X = \begin{pmatrix} x_{11} & x_{12} & \cdots \\ x_{21} & x_{22} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

$$\text{corr}(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

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EPS or PDF figures

- ▶ `\usepackage{graphicx}`
- ▶ `\includegraphics[width=0.4\textwidth]{figure}`
- ▶ Create EPS with CorelDraw, FreeHand, xfig, ...
- ▶ EPS with `latex` and `dvips`, PDF with `pdflatex`

PDF, JPG and PNG figures

- ▶ `\usepackage{pgf}`
- ▶ `\pgfdeclareimage[height=3.0cm]{figuretwo}{fancy}`
- ▶ `\pgfuseimage{figuretwo}`
- ▶ PDF from EPS using `epstopdf`, `ps2pdf`

Floating figures and tables

Input

A figure environment can carry a caption and a label.

We see in Figure 1 ...

Floating figures and tables



Figure: Image of Figure 2

We see in Figure 1 ...

Other extensions

- ▶ Hyperlinks in PDF
 - ▶ Package `hyperref`
 - ▶ Hyperlinks for all references in the text
 - ▶ Hyperlinks to external references
- ▶ PostScript tricks
 - ▶ Package `pstricks`
- ▶ Managing references in a database
 - ▶ BibTeX with style files
- ▶ Music
 - ▶ MusiXTEX package (macros plus fonts)

Presentations

- ▶ Package beamer
 - ▶ Creates a presentation PDF and handouts in one file
 - ▶ Hyperlinks between sections and to external URLs
 - ▶ Overlays

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- ▶ F. Mittelbach, M. Goossens, J. Braams, D. Carlisle, C. Rowley, *The LaTeX Companion (2nd Edition)*, Addison-Wesley, 2004.
- ▶ T. Oetiker, H. Partl, I. Hyna, E. Schlegl, *The Not So Short Introduction to L^AT_EX 2_ε*, <http://www.ctan.org/tex-archive/info/lshort/english>
- ▶ CTAN, the Comprehensive T_EX Archive Network, <http://www.ctan.org>

Summary

- ▶ L^AT_EX is a text processing program:
 - ▶ Versatile
 - ▶ Customizable
 - ▶ Extensible

Getting started...

- ▶ Practice room BBL-458.
- ▶ Use the WinShell program (in the Start menu under Tools).
- ▶ Get the exercises from the OIA website and try them.